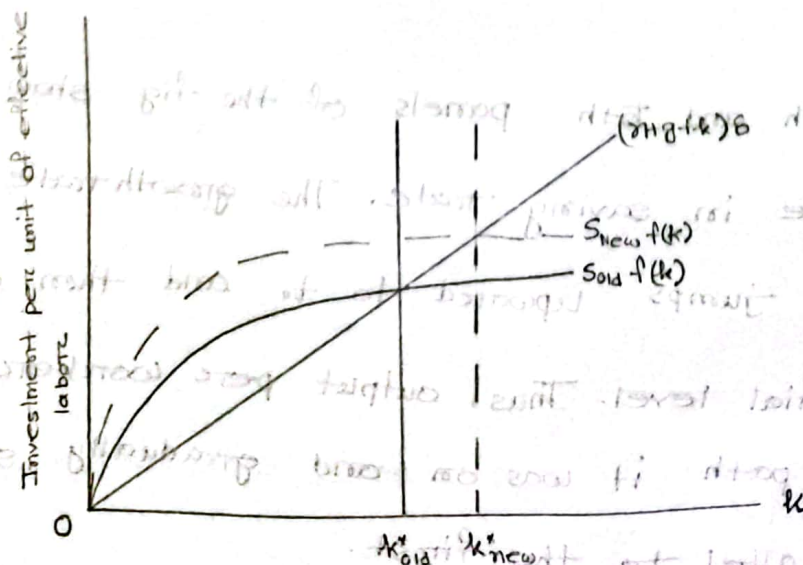


Solow Model

Show the impacts of saving ratio in the steady state solution of Solow growth model and prove that exogenous changes in the saving ratio leave only level effects on k and Y but not growth effect.

The Solow model focuses on four variables: Output (Y), Capital (k), labor (L) and knowledge or effectiveness of the labor (A). The production function is: $Y(t) = F(k(t), A(t)L(t))$; where t denotes time.

The increase in s shifts the actual investment line upward and so, k^* rises. But k doesn't immediately jump to the new value of k^* . Initially k is equal to the old value of k^* . At this level, actual investment now exceeds breakeven investment - more resources are being devoted to investment than are needed to hold k constant - and so k is positive. Thus k begins to rise. It continues to rise until it reaches the new value of k^* , at which point it remains constant.



These results are summarized in the first three panels of following fig. t_0 denotes the time of the increase in the saving rate. By assumption, s jumps up at time t_0 and remains constant thereafter. Since the jump in s causes the actual investment to exceed breakeven investment by a strictly positive amount, k jumps from zero to a strictly positive amount. k rises gradually from the old value of k^* to the new value, and k falls gradually back to zero.

Thus, a We know, $Y/L = A(k)$, when k is constant, Y/L grows at rate g , the growth rate of A . When k is increasing, Y/L grows both because A is increasing and because k is increasing. Thus its growth rate exceeds g . When k reaches the new value of k^* , however, again only the growth of A contributes to growth of Y/L and so growth rate of Y/L returns to g .

The fourth and fifth panels of the fig show how Y/L responds to the rise in saving rate. The growth rate of Y/L which is initially g , jumps up to t_0 and then gradually returns to its initial level. Thus, output per worker begins to rise above its path it was on and gradually settles into a higher path parallel to the first.

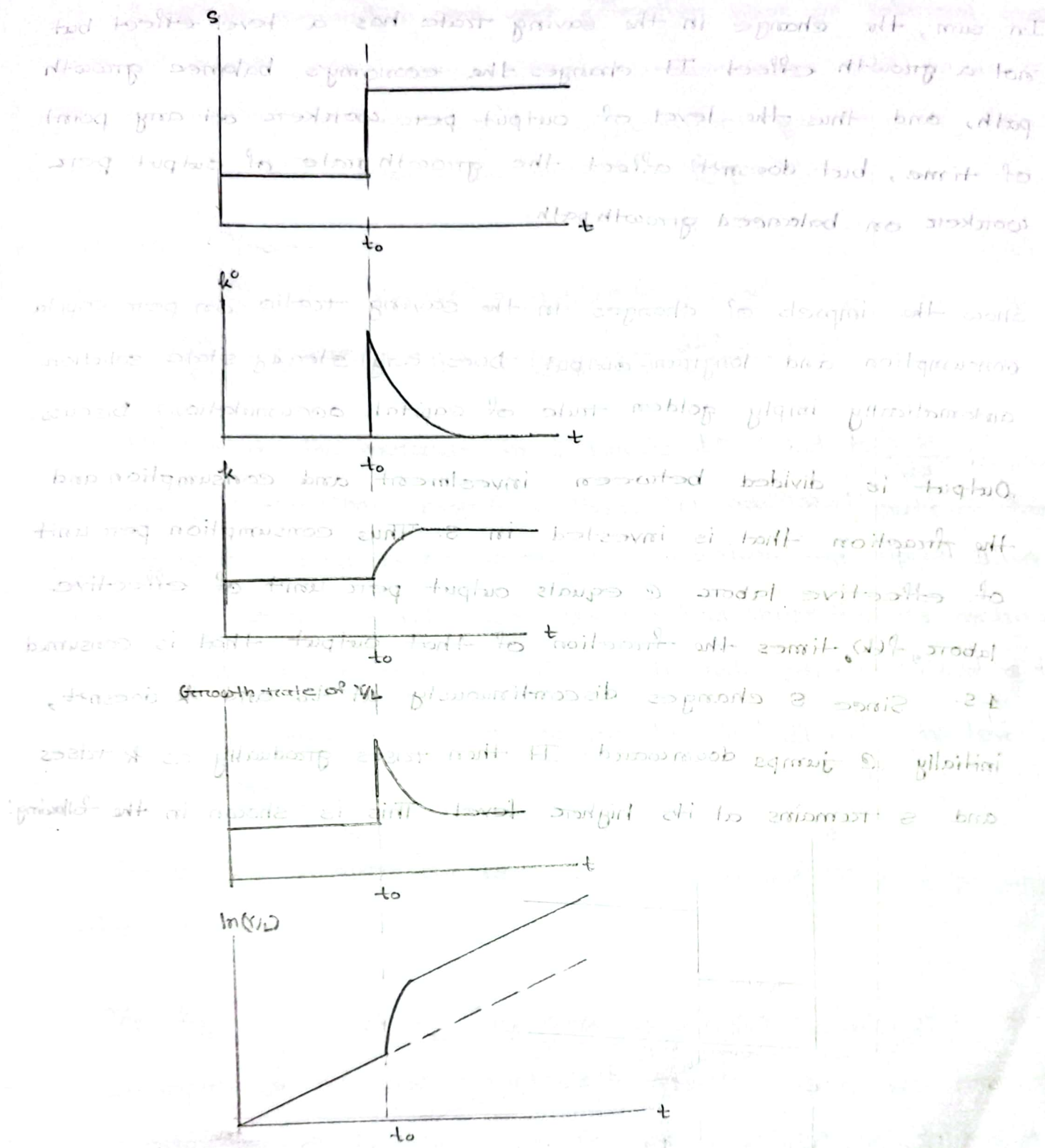
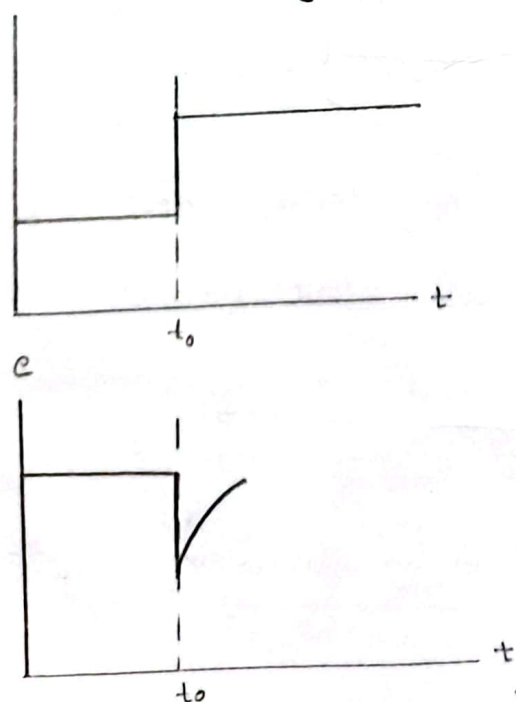


Fig: The effects of an increase in saving rate

In sum, the change in the saving rate has a level effect but not a growth effect. It changes the economy's balanced growth path, and thus the level of output per worker at any point of time, but doesn't affect the growth rate of output per worker on balanced growth path.

Show the impacts of changes in the saving ratio on per capita consumption and longrun output. Does any steady state solution automatically imply golden rule of capital accumulation? Discuss.

Output is divided between investment and consumption and the fraction that is invested is s . Thus consumption per unit of effective labor c equals output per unit of effective labor, $f(k)$, times the fraction of that output that is consumed $1-s$. Since s changes discontinuously at t_0 and k doesn't, initially c jumps downward. It then rises gradually as k rises and s remains at its higher level. This is shown in the following



Let c^* denote consumption per unit effective labor on balanced growth path. c^* equals output per unit of effective labor $f(k^*)$ minus investment per unit effective labor $sf(k^*)$. On balanced growth path, Actual investment $sf(k^*) = \text{break-even investment } (n+g+s)k^*$.

We know, $Y = c + s$

$$\therefore c^* = Y - s = Y - sf(k^*) = f(k^*) - (n+g+s)k^*$$

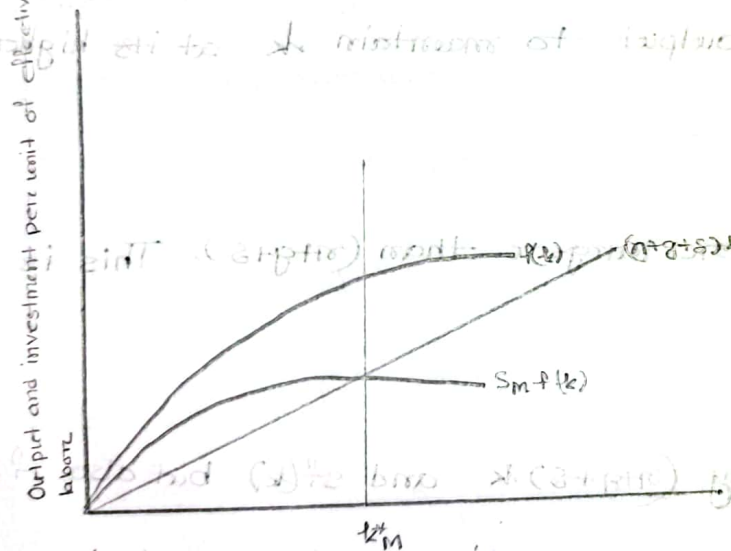
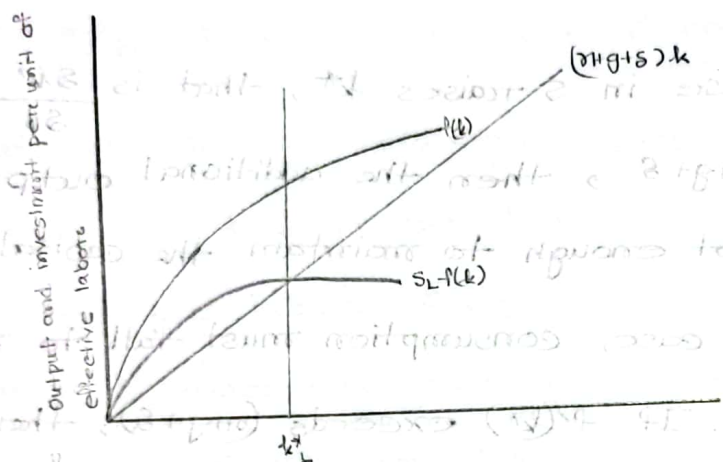
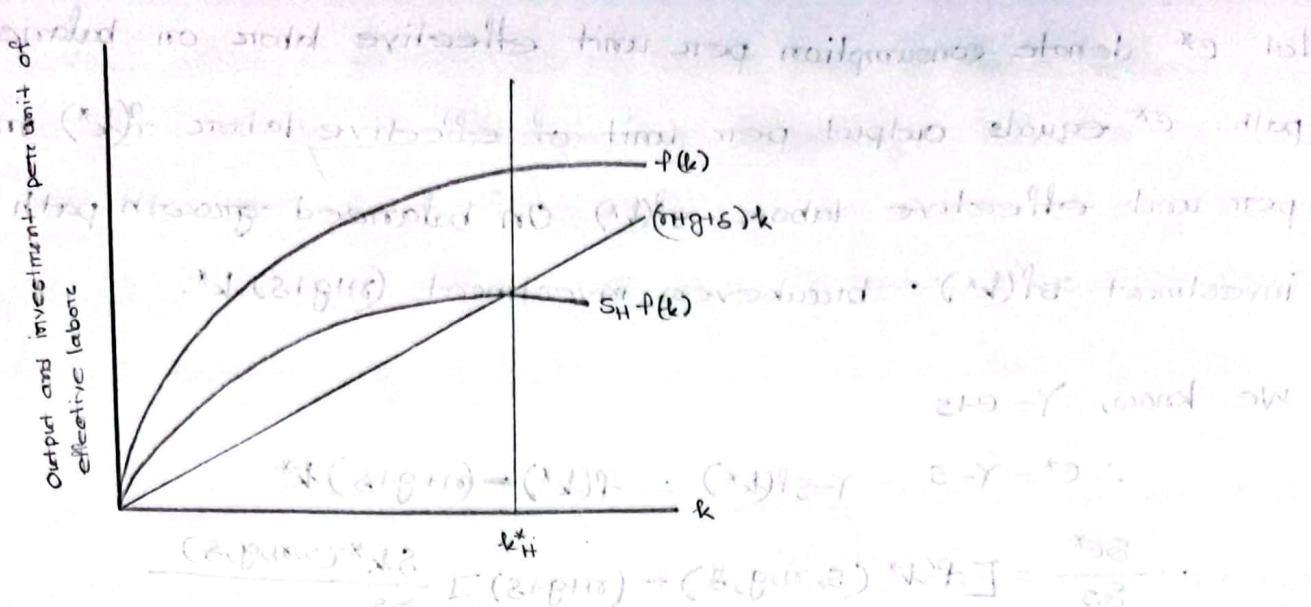
$$\therefore \frac{\partial c^*}{\partial s} = [f'(k^*)(s, n, g, \delta) - (n+g+s)] \frac{\partial k^*(s, n, g, \delta)}{\partial s}$$

We know that, the increase in s raises k^* , that is $\frac{\partial k^*}{\partial s}$ is positive.

If $f'(k^*)$ is less than $n+g+s$, then the additional output from the increased capital is not enough to maintain the capital stock at its higher level. In this case, consumption must fall to maintain the higher capital stock. If $f'(k^*)$ exceeds $(n+g+s)$, there is more than enough additional output to maintain k at its higher level and so, consumption rises.

$f'(k)$ can be smaller or larger than $(n+g+s)$. This is shown in the following fig:

This fig shows not only $(n+g+s)k$ and $sf(k)$ but also $f(k)$. Since consumption on balanced growth path equal output less break-even investment, c^* is the distance between $f(k)$ and $(n+g+s)k$ at $k = k^*$.



In the top panel, s is high and also k^* is high and $f'(k^*)$ is less than $(n+g+s)$. As a result, an increase in saving rate lowers consumption even when the economy has reached its new balanced growth path.

In the middle panel, s is low, k^* is low, $f'(k^*)$ is greater than $n+g+s$ and an increase in s raises the consumption in the long run.

Finally, in bottom panel, s is at the level that causes $f'(k^*)$ to just equal $(n+g+s)$ that is $f'(k^*)$ and $(n+g+s)k$ are parallel at $k=k^*$. In this case, a marginal change in s has no effect on consumption in the long run and consumption is at its maximum possible level among balanced growth paths. The value of k^* is known as golden rule of capital.

Find the elasticity of output per unit of effective labor on balanced growth path, y^* , with respect to the rate of population growth, n . If $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $s = 3\%$, by about how much does a fall in n from 2 percent to 1% raise y^* ?

For finding the elasticity of output per unit of effective labor, we can use the following equation:

$$\frac{n}{y^*} \frac{\partial y^*}{\partial n} = \frac{n}{(n+g+s)} \left[\frac{\alpha_k(k^*)}{1-\alpha_k(k^*)} \right]$$

Now, with $\alpha_k(k^*) = 1/3$, $g = 2\%$, and $s = 3\%$, we need to calculate the effect on y^* of a fall in n from 2% to 1%. Using the midpoint of $n = 0.015$ to calculate the elasticity gives us

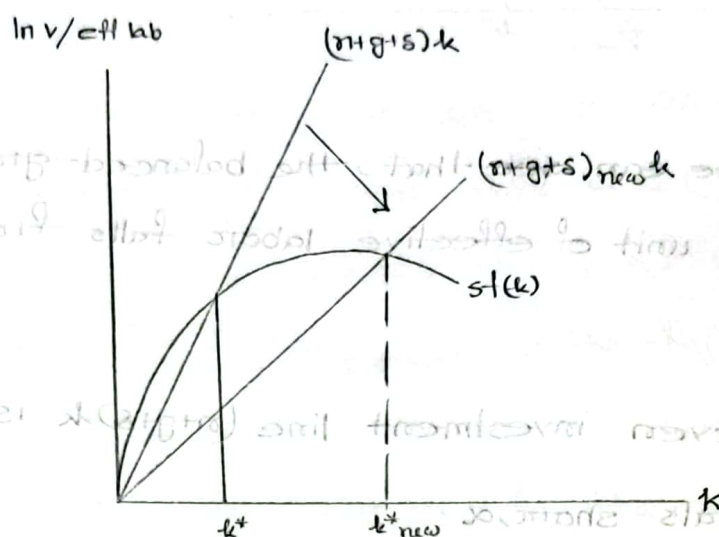
$$\frac{\gamma}{y^*} \frac{\delta y^*}{\delta n} = \frac{0.015}{0.015 + 0.02 + 0.03} \frac{(1/3)}{(1 - 1/3)} \approx -0.12$$

So, this 50% drop in the population growth rate, from 2% to 1%, will lead to approximately a 6% increase in the level of output per unit of effective labor, since $(-0.50)(-0.12) = 0.06$. This calculation illustrates the point that observed differences in population growth rates across countries are not nearly enough to account for differences in y that we see.

Describe how, if at all, each of the following developments affect the break-even and actual investment lines in our basic diagram for the Solow model:

- The rate of depreciation falls.
- The rate of technological progress rises.
- The production function is Cobb-Douglas, $f(k) = k^\alpha$ and capital's share, α rises.
- Workers exert more effort, so that output per unit of effective labor is higher than before.

(a) The slope of the break-even investment line is given by $(n+g+s)$ and thus a fall in the rate of depreciation, s , decreases the slope of the break-even investment line. The actual investment curve, $sf(k)$ is unaffected. This is shown in the following fig:

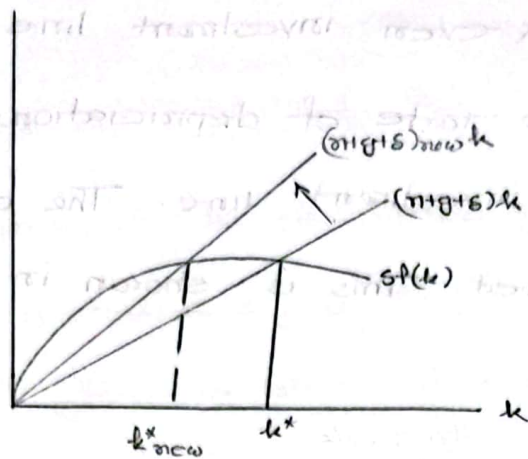


From the fig, we can see that the balanced growth path level of capital per unit of effective labor rises from k^* to k^*_{new} .

(b) Since the slope of the break-even investment line is given by $(n+g+s)$, a rise in the rate of technological progress, g , makes the break-even investment line steeper. The actual investment curve, $sf(k)$ is unaffected.

This is shown by the following fig:

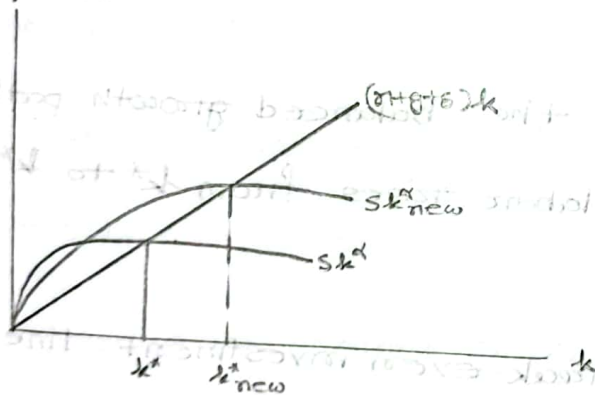
Inv/eff lab



From this fig, we can see that, the balanced-growth path of level of capital per unit of effective labor falls from k^* to k^*_{new}

(c) The break-even investment line $(n+g+s)k$ is unaffected by the rise in capital's share, α .

Inv/eff lab



The effect of a change in α on the actual investment curve, sk^{α} , can be determined by examining the derivative $\partial(sk^{\alpha})/\partial\alpha$. It is possible to show that

$$\frac{\partial sk^{\alpha}}{\partial\alpha} = sk^{\alpha} \ln k$$

For $0 < \alpha < 1$, and for possible values of k , the sign of $\partial(sk^{\alpha})/\partial\alpha$

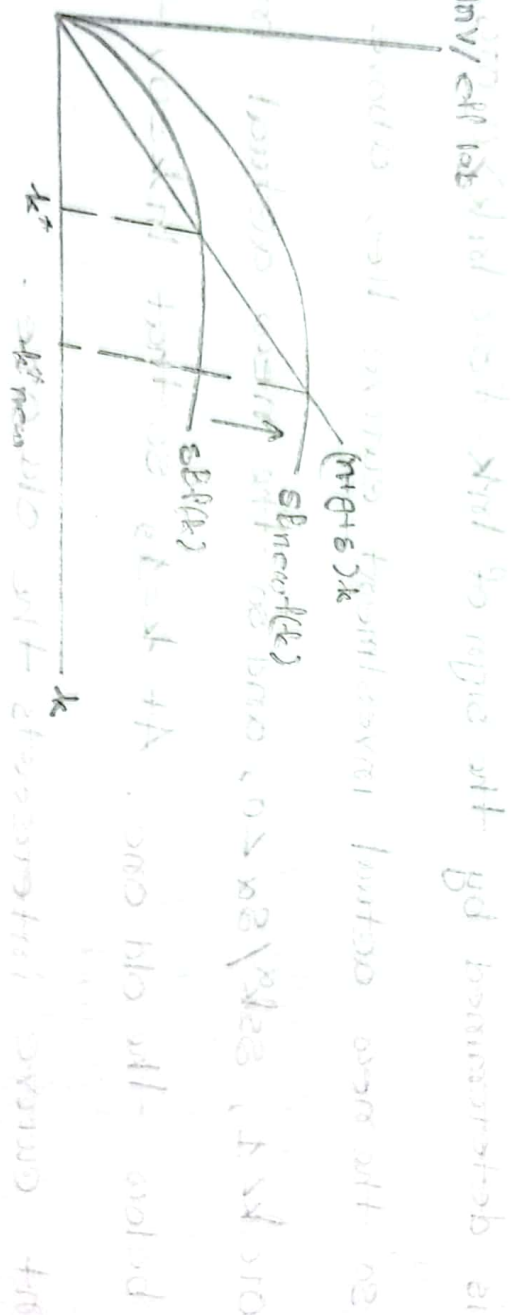
is determined by the sign of $\ln k$. For $\ln k > 0$ or $k > 1$, $s s k^\alpha / s \alpha > 0$ and so the new actual investment curve lies above the old one. For $\ln k < 0$ or $k < 1$, $s s k^\alpha / s \alpha < 0$, and so the new actual investment curve lies below the old one. At $k=1$, so that $\ln k = 0$, the new actual investment curve intersects the old one.

In addition, the effect of a rise in α on k^* is ambiguous and depends on the relative magnitudes of s and $(n+g+s)$. It is possible to show that a rise in capital's share, α , will cause k^* to rise if $s > (n+g+s)$. This case is depicted in the fig above.

(d) Let us assume that, we modify the intensive form of the production function to include a non-negative constant B , so that actual investment curve is given by $s B f(k)$, $B > 0$.

The workers exerting more effort, so that output per unit of effective labor is higher than before, can be modeled as an increase in B . This increase in B shifts the actual investment curve up. The break-even investment line $(n+g+s)k$, is unaffected.

This is shown in the following fig:



From the fig we can see that the balanced growth-path level of capital per unit of effective labor rises from k^* to k'_{new} .